torch.sparse.spdiags#

https://pytorch.org/docs/stable/generated/torch.sparse.spdiags.html

Description:

Creates a sparse 2D tensor by placing the values from rows of diagonals along specified diagonals of the output The offsets tensor controls which diagonals are set. If offsets[i] = 0, it is the main diagonal If offsets[i] < 0, it is below the main diagonal If offsets[i] > 0, it is above the main diagonal

Function Signature

```
torch.sparse.spdiags(diagonals, offsets, shape, layout=None)
→ Tensor#

Parameters

Keyword Arguments
```

Parameters

Keyword

layout (torch.layout, optional) – The desired layout of the returned tensor. torch.sparse_coo, torch.sparse_csc and torch.sparse_csr are supported. Default: torch.sparse_coo

Code Examples

```
>>> diags = torch.arange(9).reshape(3, 3)
>>> diags
tensor([[0, 1, 2],
        [3, 4, 5],
        [6, 7, 8]])
>>> s = torch.sparse.spdiags(diags, torch.tensor([0, -1, -2]),
(3, 3))
>>> s
tensor(indices=tensor([[0, 1, 2, 1, 2, 2],
                       [0, 1, 2, 0, 1, 0]]),
       values=tensor([0, 1, 2, 3, 4, 6]),
       size=(3, 3), nnz=6, layout=torch.sparse_coo)
>>> s.to_dense()
tensor([[0, 0, 0],
        [3, 1, 0],
        [6, 4, 2]])
```

```
>>> diags = torch.arange(9).reshape(3, 3)
>>> diags
tensor([[0, 1, 2],[3, 4, 5], [6, 7, 8]])
>>> s = torch.sparse.spdiags(diags, torch.tensor([0, -1, -2]),
(3, 3), layout=torch.sparse_csr)
>>> s
tensor(crow_indices=tensor([0, 1, 3, 6]),
       col_indices=tensor([0, 0, 1, 0, 1, 2]),
       values=tensor([0, 3, 1, 6, 4, 2]), size=(3, 3), nnz=6,
       layout=torch.sparse_csr)
>>> s.to_dense()
tensor([[0, 0, 0],
        [3, 1, 0],
        [6, 4, 2]])
```

```
>>> diags = torch.tensor([[1, 2], [3, 4]])
>>> offsets = torch.tensor([0, -1])
>>> torch.sparse.spdiags(diags, offsets, (5, 5)).to_dense()
tensor([[1, 0, 0, 0, 0],
        [3, 2, 0, 0, 0],
        [0, 4, 0, 0, 0],
        [0, 0, 0, 0, 0],
        [0, 0, 0, 0, 0]])
```

```
>>> diags = torch.tensor([[1, 2, 3], [1, 2, 3], [1, 2, 3]])
>>> torch.sparse.spdiags(diags, torch.tensor([0, 1, 2]), (5,
5)).to_dense()
tensor([[1, 2, 3, 0, 0],
        [0, 2, 3, 0, 0],
        [0, 0, 3, 0, 0],
        [0, 0, 0, 0, 0],
        [0, 0, 0, 0, 0]])
```

Notes & Warnings

NOTE Note When setting the values along a given diagonal the index into the diagonal and the index into the row of diagonals is taken as the column index in the output. This has the effect that when setting a diagonal with a positive offset k the first value along that diagonal will be the value in position k of the row of diagonals

Detailed Documentation

Documentation

Creates a sparse 2D tensor by placing the values from rows of diagonals along specified diagonals of the output

The offsets tensor controls which diagonals are set.

If offsets[i] = 0, it is the main diagonal

If offsets[i] < 0, it is below the main diagonal

If offsets[i] > 0, it is above the main diagonal

The number of rows in diagonals must match the length of offsets, and an offset may not be repeated.

diagonals (Tensor) – Matrix storing diagonals row-wise

offsets (Tensor) – The diagonals to be set, stored as a vector

shape (2-tuple of ints) – The desired shape of the result

layout (torch.layout, optional) – The desired layout of the returned tensor. torch.sparse_coo, torch.sparse_csc and torch.sparse_csr are supported. Default: torch.sparse_coo

Set the main and first two lower diagonals of a matrix:

Change the output layout:

Set partial diagonals of a large output:

When setting the values along a given diagonal the index into the diagonal and the index into the row of diagonals is taken as the column index in the output. This has the effect that when setting a diagonal with a positive offset k the first value along that diagonal will be the value in position k of the row of diagonals

Specifying a positive offset: